21ECP401L-MAJOR PROJECT

Intelligent Radio Resource Allocation for 5G Traffic using Federated Deep Reinforcement learning with Long Short-Term Memory

A PROJECT REPORT

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In partial fulfillment of the requirements for the Degree

of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS & COMMUNICATION ENGINEERING

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(Deemed to be University u/s 3 of UGC Act, 1956)

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(Under Section 3 of UGC Act, 1956)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that the project report titled **“Intelligent Radio Resource Allocation for 5G Traffic using Federated Deep Reinforcement learning with Long Short- Term Memory”,** submitted for **21ECP401L – Major Project,** is a Bonafide work carried out by the following students**:**

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in partial fulfilment of the requirements for the award of the degree of **B.Tech in Electronics and Communication Engineering**. This work has been carried out under my supervision and guidance. To the best of my knowledge, the work reported herein has not been submitted, either in part or in full, for the award of any other degree or diploma by this or any other candidate.

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Signature of the Internal Examiner **Signature of the External Examiner**

DECLARATION

We hereby declare that the Major Project entitled "**Intelligent Radio Resource Allocation for 5G Traffic using Federated Deep Reinforcement learning with Long Short-Term Memory**" to be submitted for the Degree of Bachelor of Technology is our original work as a team and the dissertation has not formed the basis of any degree, diploma, associateship or fellowship of similar other titles. It has not been submitted to any other University or institution for the award of any degree or diploma.

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ABSTRACT

The increasing demand for high-speed and reliable communication in 5G networks has introduced multiple types of traffic, such as eMBB, URLLC, and mMTC, each requiring different levels of service quality. Managing radio resources efficiently in such a ever-changing environment has become a major problem, as conventional methods are mostly fixed and respond only after changes occur. This often leads to inefficient spectrum usage and inconsistent network performance. To overcome these limitations, this project presents an intelligent radio resource allocation approach using Federated Deep Reinforcement Learning. The system combines different techniques to make better and faster decisions. LSTM models are used to understand and predict traffic patterns over time, while Graph Neural Networks help capture relationships between different network nodes. A distributed Deep Reinforcement Learning setup allows each base station to make real-time decisions, and federated learning enables collaboration between them without sharing raw data. A simulated 5G dataset is used to train and test the proposed system under realistic conditions. The performance is evaluated using parameters such as latency, throughput, fairness, and energy efficiency. These results shows improved overall network performance as well as adapting well to changing conditions. This makes it a suitable solution for future 5G and beyond communication systems, where intelligent and efficient resource management is essential.

Sustainable Development Goal

SDG 8 – Decent Work and Economic Growth, SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities

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LIST OF ABBREVIATIONS

Abbreviation	**Meaning**
5G	Fifth Generation
AI	Artificial Intelligence
DDQN	Double Deep Q-Network
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
eMBB	Enhanced Mobile Broadband
FDRL	Federated Deep Reinforcement Learning
GNN	Graph Neural Network
IoT	Internet of Things
k-NN	k-Nearest Neighbor
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
ML	Machine Learning
mMTC	Massive Machine Type Communication
QoS	Quality of Service
RAN	Radio Access Network
RMSE	Root Mean Square Error
RRM	Radio Resource Management
SDN	Software Defined Networking
SINR	Signal-to-Interference-plus-Noise Ratio
URLLC	Ultra-Reliable Low Latency Communication

LIST OF SYMBOLS

Symbol	Meaning
$Q_{\pi}(s, a)$	Action-value function
$Q^{*}(s, a)$	Optimal Q-function
Π	Policy
s	State
a	Action
r	Reward
$\lambda \max_{a'}$	Discount factor
ω	Neural network weights
ω^{-}	Target network parameters
ω_{g}	Global model parameters
ω_{i}	Local model parameters
η_{i}	Number of samples at node
X'	Node feature matrix (GNN)
$\hat{A}$	Adjacency matrix
σ	Activation function
$\hat{X}_{s}(t + 1)$	Predicted traffic load
N	Number of samples/packets
$t_{send_{i}}$	Packet send time
$t_{receive_{i}}$	Packet receive time
B	Bandwidth
P_{sent}	Packets sent
$P_{received}$	Packets received
E	Epsilon

CHAPTER 1

INTRODUCTION

1.1 Introduction

Wireless communication has evolved significantly from 1G to 5G, with each generation bringing major improvements in capability and usage. 1G systems were limited to analog voice communication with low quality and security. 2G introduced digital communication, enabling clearer voice calls along with SMS services. With 3G, mobile networks moved beyond voice, offering internet access, multimedia messaging, and basic data services. 4G further transformed connectivity by providing high-speed data, enabling applications such as HD video streaming, online gaming, and mobile cloud services. However, as the number of connected devices and data demand grew rapidly, 4G began to face limitations in handling massive connectivity, ultra-low latency, and diverse service requirements.

This led to the development of 5G, which is not just an upgrade in speed but a complete shift in network design. 5G is built to support a wide range of advanced applications, including smart cities, Internet of Things (IoT), autonomous vehicles, remote healthcare, and real-time immersive services. Unlike previous generations that focused mainly on human communication, 5G also targets machine-to-machine communication and intelligent automation. A key feature of 5G is network slicing, which allows a single physical network to be divided into multiple virtual networks, each optimized for specific types of services. The three primary slices are enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communication (URLLC), and massive Machine-Type Communication (mMTC).

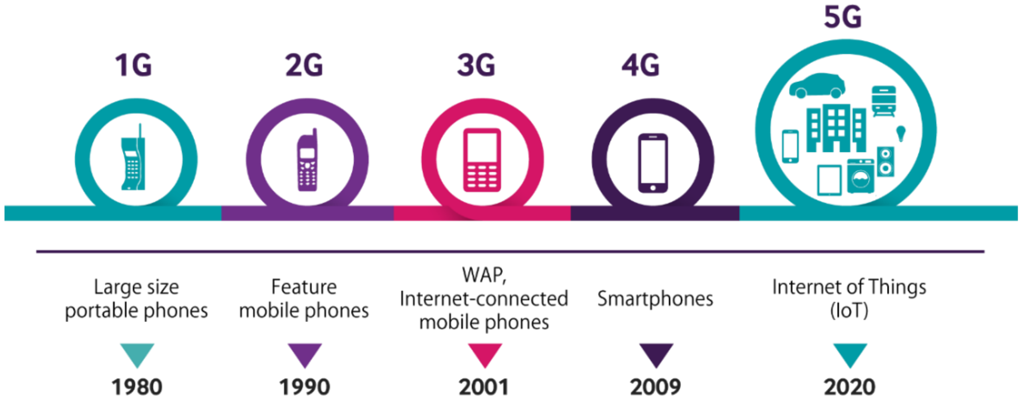

Figure 1.1: Evolution of 1G to 5G

The eMBB slice supports high data rate applications like video streaming, augmented and virtual reality, URLLC focuses on ultra-low latency and high reliability for critical tasks such as autonomous driving and remote surgery and mMTC enables large-scale IoT connectivity with low power and data requirements, creating conflicting performance requirements that complicate radio resource management. Conventional static or rule-based scheduling approaches struggle to mange with rapidly changing traffic patterns, dynamic channel conditions, and the increasing density of the 5G services, thereby motivating the development of intelligent and adaptive resource allocation mechanisms.

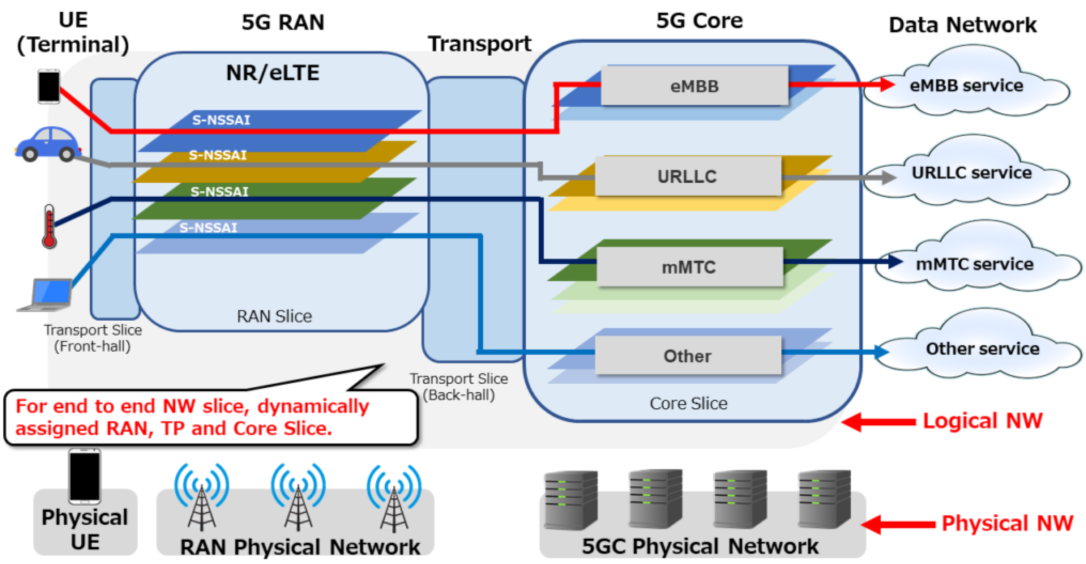

Figure 1.2: 5G Technology Topology

To address these challenges, 5G networks introduce advanced concepts such as network slicing and RAN scheduling, enabling multiple logical services to coexist over a shared physical infrastructure. In RAN slicing, PRBs are dynamically allocated across network slices according to services demands and QoS constraints. However, the coexistence of URLLC and eMBB traffic introduces new challenges in scheduling; due to the need for immediate URLLC traffic, they can negatively impact eMBB throughput and overall spectral efficiency if not managed effectively. Consequently, designing intelligent scheduling and resource allocation strategies capable of balancing latency, reliability, and throughput remains a key research problem in next-generation wireless networks.

There have been many breakthroughs in recent times with regard to the application of AI and ML techniques in improving resource allocation in radios. The use of LSTM neural networks in predicting traffic helps make better allocation decisions in advance due to accurate predictions of the requirements in various slices of the network. Also, GNN provides modeling of the network topologies due to its ability to model the spatial relationship between the nodes and their interference. Reinforcement learning methods such as DRL are useful in solving resource allocation problems.

Though the benefits have been offered, there are some scalability issues, privacy problems, and excessive overhead for communication associated with machine learning techniques that are based on centralization. This is when Federated Learning comes into play, which serves as a great decentralized technique to help multiple network nodes to train a global model in a cooperative way without actually sharing any data and ensuring privacy along with improved efficiency in learning. It also facilitates cooperative decision-making among the nodes as far as efficient allocation of resources is concerned.

In response to the problems described above, this research introduces a slicing aware radio resource allocation mechanism for 5G and beyond wireless systems. The suggested approach includes LSTM-based traffic prediction, which would ensure pro-active analysis of demands; GNN-based modeling, which would incorporate information about the structure of the network and make it possible to coordinate decisions in different cells; and federated multi-agent DDQN based reinforcement learning, which will allow for efficient resource allocation while preserving privacy. The suggested system should address the need for fast URLLC operation through intelligent puncturing and scheduling without sacrificing the quality of the experience of eMBB service users.

1.2 Objective

- To analyze real 5G traffic data and characterize slice-level demand for eMBB, URLLC and mMTC services.
- To develop a predictive radio resource management framework using LSTM and DNN models based on traffic characteristics.
- To enable intelligent, slice-aware radio resource allocation using Federated Deep Reinforcement Learning for scalable 5G networks.

1.3 Overview of Radio Resource Management in 5G

In Fifth Generation (5G) mobile networks, the definition of Radio Resource Management (RRM) is the process by which limited mobile network resources are allocated to multiple end users and services falling under a variety of performance criteria. The analysis and allocation of wireless resources become increasingly complex due to the concurrent degradation of three different classes of 5G traffic (eMBB, URLLC, and mMTC), which leads to simultaneous satisfaction of multiple objectives.

Traditional methods of RRM are not able to adapt quickly enough to the complexities of 5G networks and to operate optimally within dynamic environments. Intelligent approaches based on machine learning algorithms will provide an optimal level of resource allocation through predictive and adaptive resource allocation.

1.4 Need for Intelligent Resource Allocation

The growing need for fast, dependable, and low-latency communication is resulting in higher demands on resource allocation than ever. Resource-based assignments are challenged due to congestion, interference, and fluctuating user needs, making it hard to achieve peak performance with traditional resources.

An intelligent system predicts a future network system and allocate resources accordingly. These results are then applied to improved performance through greater efficiency, shorter delays, and a more complete user experience.

1.5 Proposed System Overview

The proposed system combines the LSTM traffic prediction, GNN network state representation, and DRL decision-making in a federated learning architecture. The primary purpose of each element is to enhance resource allocation.

By using an integrated approach that uses these supportive models, the proposed framework can adapt to the dynamic nature of the given network while optimizing resource utilization in real time. This provides a scalable, effective means of meeting the future needs for wireless communications systems.

CHAPTER 2

LITERATURE SURVEY

2.1 Traffic Analysis and Prediction

- **D. Kulkarni, M. Venkatesan, and A. V. Kulkarni, "Deep learning traffic prediction and resource management for 5G RAN slicing," Wireless Personal Communications, 2022.**

  Kulkarni et al. proposed a CNN–LSTM-based model to predict traffic demand for proactive resource allocation in 5G slicing. The model achieves high prediction accuracy (~99%) and improves allocation efficiency by ~6–8%. However, it relies on heuristic optimization and lacks reinforcement learning and spatial awareness, limiting adaptability in dynamic environments.

- **F. Lotfi and F. Afghah, "Open RAN LSTM traffic prediction and slice management using deep reinforcement learning," Proc. IEEE GLOBECOM, 2022.**

  Lotfi and Afghah combined LSTM-based traffic prediction with DRL for slice management in Open RAN. The approach improves QoS and resource utilization by enabling proactive scheduling under traffic variations. However, it does not include federated learning or spatial topology modelling, reducing scalability in multi-cell deployments.

2.2 Topology-Aware Resource Allocation

- **T. Perera, S. Atapattu, Y. Fang, P. Dharmawansa, and J. Evans, "Flex-Net: A graph neural network approach to resource management in flexible duplex networks," IEEE Transactions on Wireless Communications, 2023.**

  Perera et al. proposed a GNN-based model to capture spatial interference and network topology for optimized resource allocation. The model achieves near-optimal performance (~95.8%) with efficient computation. However, it does not support network slicing or QoS differentiation across traffic types.

- **H. Wang, Y. Wu, G. Min, and W. Miao, "A graph neural network-based digital twin for network slicing management," IEEE Internet of Things Journal, 2023.**

  Wang et al. introduced a GNN-based digital twin model to represent real-time network behaviour and topology. The system improves monitoring accuracy and decision support for slice management. However, it lacks reinforcement learning integration and does not include traffic prediction for dynamic optimization.

2.3 Intelligent Decision-Making

- **R. M. Sohaib, Y. Sambo, O. Onireti, R. Swash, S. Ansari, and M. A. Imran, "Intelligent resource management for eMBB and URLLC in 5G and beyond wireless networks," IEEE Communications Surveys & Tutorials, 2023.**

Sohaib et al. proposed a DRL-based framework to balance throughput and latency for eMBB and URLLC traffic. The system achieves URLLC reliability up to 10^{-6}while maintaining eMBB throughput in the range of 20–60 Mbps. However, it is centralized and lacks spatial mode ling, limiting scalability.

- **G. Chen, R. Shao, F. Shen, and Q. Zeng, "Slicing resource allocation based on dueling DQN for eMBB and URLLC hybrid services," Sensors, 2023.**

Chen et al. proposed a Dueling-DQN model that improves decision stability by separating value and advantage functions. The approach enhances network utility by ~8–11% and achieves near 100% QoE. However, it is limited to single-cell scenarios and lacks distributed learning.

- **D. Shome and A. Kudeshia, "Deep Q-learning for 5G network slicing with diverse resource stipulations and dynamic data traffic," Proc. IEEE AIIC, 2021.**

Shome and Kudeshia developed a DQN-based resource allocation model for dynamic slicing. The approach improves flexibility compared to static methods but suffers from instability and Q-value overestimation. It also lacks advanced architectures and federated learning support.

2.4 Distributed Learning using Federated Learning

- **Noman, K. Dimyati, K. A. Noordin, E. Hanafi, and A. Abdrabou, "FeDRL-D2D: Federated deep reinforcement learning empowered resource allocation scheme for energy efficiency maximization in D2D-assisted 6G networks," IEEE Access, 2023.**

Noman et al. proposed a federated DDQN-based framework for distributed resource allocation. The model achieves energy efficiency gains of 16–73% and improves system throughput by up to 47% while preserving data privacy. However, it focuses on D2D scenarios and does not address slice-aware QoS requirements or traffic prediction.

- **H. Zhuo, W. Feng, Y. Lin, Q. Xu, and Q. Yang, "Federated deep reinforcement learning for distributed resource allocation," IEEE Transactions on Neural Networks and Learning Systems, 2022.**

  Zhuo et al. proposed a federated DRL framework enabling collaborative learning across distributed agents. The model reduces communication overhead and improves scalability while maintaining privacy. However, it is not slice-aware and lacks QoS-driven optimization and prediction mechanisms.

- **Filali, Z. Mlika, S. Cherkaoui, and A. Kobbane, "Dynamic SDN-based radio access network slicing with deep reinforcement learning for URLLC and eMBB services," IEEE Transactions on Network and Service Management, 2022.**

  Filali et al. proposed a multi-agent DDQN-based SDN framework for dynamic slicing. The system converges within ~2000 episodes and maintains URLLC latency below 10 ms. However, it relies on centralized control and lacks predictive traffic modelling.

2.5 Research Gap

The literature reviewed has established that the majority of prior studies have been focused on different areas of radio resource management including DRL-based resource allocation, LSTM-based traffic forecasting, GNN-based spatial modelling, and federated learning. Each of these areas has produced results that generally improve resource management in that category, however the majority have done so independently, and therefore do not provide an integrated way of addressing all of the challenges presented by dynamic traffic conditions, interference and multi-slice QoS requirements in the 5G network.

A number of models have been created using a centralized learning approach which limits scalability and increases communication costs. Additionally, some models do not include time-based predictions or consider the physical layout of the network to help provide an understanding of how best to deliver resources to the end user, therefore limiting their ability to function effectively in dynamic and multi-cell environments. Furthermore, the lack of slice-aware optimization in many studies fails to balance the needs of eMBB, URLLC and mMTC services.

The need for an integrated framework that incorporates traffic forecasting, topology-sensitive modelling and distributed learning is evident as a means of overcoming the limitations identified. The proposed research provides an integrated solution by combining LSTM for time-based predictions, GNN for spatial feature extraction and Federated DDQN for the provision of adaptive and privacy-preserving RRM in order to efficiently scale RRM implementations in future 5G systems.

CHAPTER 3

SYSTEM MODEL

3.1 Introduction to 5G Proposed Model

The proposed model presents an intelligent and adaptive RRM framework that integrates 5G technology into existing 5G networks in order to provide effective and efficient dynamic allocation of radio resources across the three main categories of 5G traffic: eMBB, uRLLC, and mMTC. Each traffic slice has unique Qos requirements for the intelligent allocation mechanism. In addition, the combined use of LSTM and FDRL with DQN agent will allow the RRM framework to quickly adapt to changing network conditions while providing significant performance improvements and efficient management of multi-slice 5G network infrastructures.

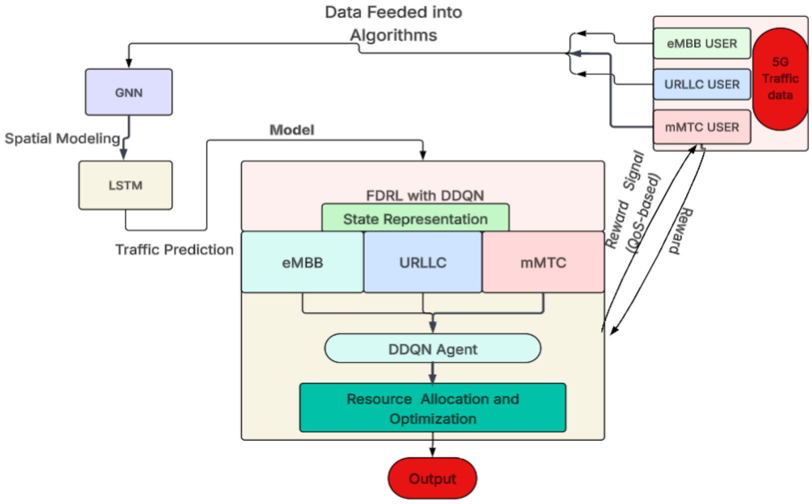

Figure 3.1: Architecture of the 5G proposed model

From the system model shown in Figure 3, the system has three main phases. Firstly, the data set is analyzed and preprocessed, after which it is fed into the machine learning models such as the GNN used for spatial analysis and the LSTM network for traffic prediction. After the data been predicted and analyzed, it is then trained to the Federated DDQN model to make smarter decisions and allocate resources effectively. Lastly, the performance is evaluated through Quality of Service (QoS) measures, hence ensuring efficient AI-based management and optimization of 5G network resources.

3.2 Input Data and Preprocessing

The system uses a synthetic 5G QoS dataset generated and processed in a Python environment. The dataset includes key network parameters that help in analyzing and optimizing resource allocation.

Time(ms)

In 5G network, time refers to the amount of time it takes to process or send the packet, which is measured in milliseconds. The larger the value of time, the longer it will take to process the packet. On the other hand, the smaller values show faster processing time and less delay.

Packet Size

The packet size indicates the amount of information can be fit inside a single packet. Large packets are generally used for high bandwidth applications like video streaming, while small packets are used in applications that require low latency.

SINR

The SINR evaluates the quality of wireless signals by comparing the level of the signal against the interference and noise in total. Higher the values of SINR better is the signal strength and higher the performance during data transfer with lower error rates.

Queue Length

Queue length refers to the number of packets in line in the buffer waiting for transmission. Packet's lengths which are too high indicates congestion in the network and may lead to delays and loss of packets. So by managing these queue lengths results in a good network performance.

Purpose of Preprocessing:

- Remove noise and inconsistencies
- Normalize data
- Convert the data into a serial form that is appropriate for LSTM models.
- Design a state description for reinforcement learning.

The data set reflects the practical nature of different 5G traffic types with respect to their QoS features. The eMBB traffic type is marked by high throughput and strong burstiness due to bandwidth-demanding applications. URLLC traffic focuses on low latency and reliable communication, making it suitable for real-time services. In contrast, mMTC traffic exhibits low data rates and stable behavior, as it is primarily used for IoT-based applications with periodic data transmission.

3.3 GNN -Spatial Modeling

The GNN models the network topology and interference relationships between different cells. It represents the base strations and users as nodes and captures connections to learn the interference patterns. The model shows the relationships patterns of the three slices using the KNN PCA graph. Let X denote node features representing base station load and channel conditions, and $\hat{A}$ be the adjacency matrix describing network connectivity. The graph convolution operation is defined as:

$$X' = \sigma(\hat{A}XW) \tag{3.1}$$

- X'= updated node feature matrix after convolution.
- $\hat{A}$ = normalized adjacency matrix.
- X = input node feature matrix.
- W = learnable weight parameters of the model.
- σ = nonlinear activation function applied after aggregation.

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The GNN captures spatial correlations to enable coordinated multi-cell optimization.

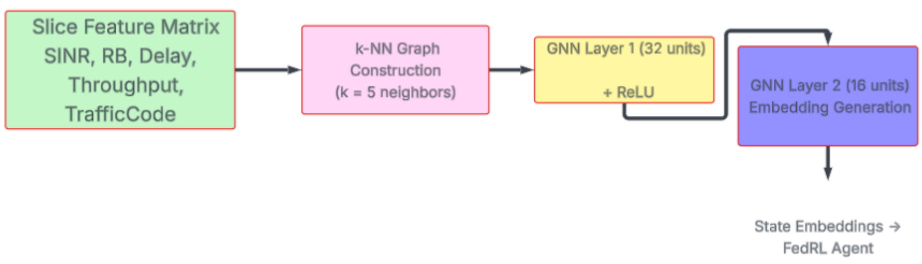

Figure 3.2: Architecture of the GNN model

3.4 LSTM for Throughput Traffic Predictions

Long Short-Term Memory network is used to predict future traffic throughput by learning patterns from past data. The input consists of packet sizes of each slice varying in the data. These inputs are first organized into fixed-length sequences (e.g., 20-time steps), allowing the model to capture short-term traffic trends instead of relying on single observations. The application of this methodological approach allows for a deeper understanding of the dynamics of the changing network environment, which is necessary in dynamic 5G networks. The sequence is fed into the LSTM layer that has 128 nodes and uses memory cells and gates to store the relevant history data and model temporal changes. The output from this layer is passed to the dense layer to predict the throughput at the next step.

This prediction can be used to make proactive resource allocation decisions in slice-aware networks.

To improve the proactive resource allocation, LSTM networks are employed to predict future traffic demands for each network slice. With the given historical traffic data H_t, the predicted load for the next time step is:

$$\hat{\lambda}_s(t+1) = f_{LSTM}(H_t) \tag{3.2}$$

- $\hat{\lambda}_s(t+1)$ = the predicted traffic load of slice $s$at the next time step.
- f_{LSTM} = the LSTM model used for prediction.
- H_t = historical traffic data up to time t.

This prediction enables anticipatory resource scheduling before congestion occurs.

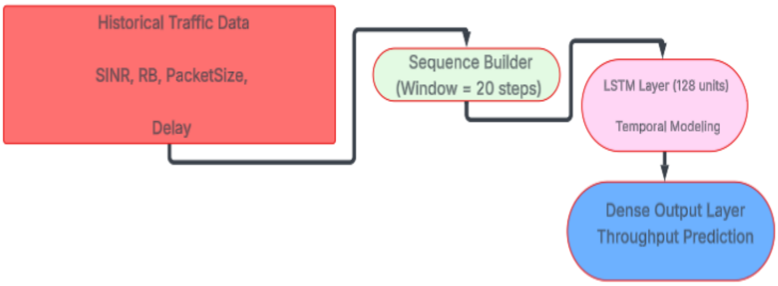

Figure 3.3: Architecture of the LSTM Model

3.5 FDRL with DDQN agent

The FDRL model is the core decision-making block for the proposed model. It enables distributed and private resource allocation over many 5G slices without sharing raw data. Several agents perform local learning and periodically aggregate model parameters to achieve coordinated optimization. This decentralized framework is perfect for multi-slice systems such as eMBB, URLLC, and mMTC, where independent network parts must cooperate while making scalable judgments.

The framework collects slice-specific network states, including traffic demand, resource block availability, queue status, and SINR. These slice-level observations are aggregated while retaining slice awareness and sent to a Slice-Aware Double Deep Q-Network (DDQN) agent. This DDQN agent approaches an adaptive and effective slice-aware resource allocation mechanism for next-generation networks through ongoing feedback and parameter modifications.

Figure 3.4: Architecture of the FDRL with DDQN Model

3.5.1 DRL Formulation

The objective of reinforcement learning is to learn an optimal policy that maximizes the expected cumulative reward. The action-value function is defined as:

$$Q_\pi(s,a) = E[G_t \mid s,a] \tag{3.3}$$

- $Q_\pi(s,a)$ = expected return when taking action $a$in state $s$under policy π.
- G_t = cumulative future reward starting at time t.
- s = current state of the environment.
- a = action chosen by the agent.

The optimal Q-function follows the Bellman optimality equation:

$$Q^*(s,a) = E[r + \lambda \max_{a'} Q^*(s',a')] \tag{3.4}$$

- $Q^*(s,a)$ = optimal Q-function.
- r = immediate reward received after taking action a.
- $\lambda \max_{a'}$ = discount factor, controls the importance of future rewards.
- s' = next state after taking action.
- a' = possible future actions.

Due to large state and action spaces, Deep Q-Networks (DQN) are used to approximate the Q-function:

$$Q(s,a;\omega) \tag{3.5}$$

where:

- $Q(s, a; \omega)$ = approximated Q-value using a neural network.
- ω = neural network parameters (weights).

where ωrepresents the neural network parameters.

The loss function used for training is:

$$L(\omega) = E[(y - Q(s, a; \omega))^2] \tag{3.6}$$

- $L(\omega)$ = loss function used to update the network.
- y = target value.
- The expression measures the difference between predicted and target Q-values.

where the target value is given by:

$$y = r + \lambda \max_{a'} Q(s', a'; \omega^-) \tag{3.7}$$

- ω^- = parameters of the target network, which are updated less frequently.
- r, $\lambda \max_{a'}$, s', and a'retain their earlier meanings.

3.5.2 Double Deep Q-Network

The standard DQN suffers from overestimation bias because it uses the same network for both action selection and evaluation. To overcome this issue, DDQN is used.

DDQN target value is computed as:

$$y = r + \lambda Q\left(s', \arg\max_{a'} \quad Q(s', a'; \omega)\,; \omega^-\right) \tag{3.8}$$

- The action selection ($\arg\max_{a'} Q(s', a'; \omega)$) is done using the main network.

In this approach:

- The main network **(ω)** selects the best action
- The target network **(ω^-)** evaluates that action

This separation improves training stability and reduces overestimation, leading to better convergence.

3.5.3 Federated Learning Integration

To ensure privacy and scalability, federated learning is incorporated into the DRL framework. Each base station trains a model using the previous data and time to time shares only the model parameters.

The global model is then updated using the weighted aggregation:

$$\omega_g = \frac{\sum_{i=1}^{K} \eta_i \ \omega_i}{\sum_{i=1}^{K} \eta_i} \tag{3.9}$$

where:

- ω_i= local model parameters
- η_i= number of training samples at node i
- K= total number of nodes

The updated global model is then distributed back to all agents for further training. This approach reduces communication overhead and preserves data privacy while enabling collaborative learning.

3.6 Resource Allocation Module

This module is where Resource Learning Decision is done based on Allocation Strategy:

- URLLC: Highest priority (low latency)
- eMBB: Throughput optimization
- mMTC: Efficient low-power allocation

Thus, this approach ensures Qos differentiation and balanced performances across slices and thus make efficient decision-making learning for the model.

3.7 Feedback and Reward Calculation

The performance metrics used for evaluating the data of the proposed model are as follows:

Throughput

The throughput refers to the total amount of data successfully transmitted over the network per unit time.

$$\text{Throughput} = \frac{\sum_{i=1}^{N} \text{Data}_i}{\sum_{i=1}^{N} \text{Time}_i} \tag{3.10}$$

where:

- ω_g = global model parameter set.
- ω_i = local model parameters from node i.
- η_i = number of training samples used at node i.
- K = total number of participating nodes (base stations).

Here Data_i denotes the data transmitted in the i^{th}instance, Time_irepresents the corresponding transmission time, and $N$is the total number of samples. Throughput is evaluated by considering the combined performance of eMBB, URLLC, and mMTC traffic slices, and the average value reflects the overall data transmission capability of the network.

Average Latency

The latency is the average time delay between sending and receiving data packets.

$$\text{Latency} = \frac{1}{N}\sum_{i=1}^{N}(t_{receive_i} - t_{send_i}) \tag{3.11}$$

- t_{send_i} = sending time of the i^{th}packet.
- $t_{receive_i}$ = receiving time of the i^{th}packet.
- N = total number of packets.

Latency in the proposed system is computed as a combined average across all slices, showcasing the overall delay performance of the network.

Packet Loss

The packet loss indicates the percentage of data packets that have been lost during transmission.

$$\text{Packet Loss (\%)} = \frac{P_{sent} - P_{received}}{P_{sent}} \times 100 \tag{3.12}$$

- P_{sent} = total no. of packets sent.
- $P_{received}$ = no. of packets successfully received.

Packet loss is measured across all slices to evaluate the overall reliability of the system.

Spectral Efficiency

Spectral efficiency shows how efficiently the available bandwidth is utilized for data transmission.

$$\text{Spectral Efficiency} = \frac{\text{Throughput}}{B} \tag{3.13}$$

Where, $'B'$ is channel bandwidth in Hz. Spectral efficiency is also using the combined throughput of all slices, indicating efficient utilization of available spectrum.

Energy Efficiency

Energy efficiency refers to the amount of data transmitted per unit of energy consumed.

$$\text{Energy Efficiency} = \frac{\text{Throughput}}{\sum_{i=1}^{N} \text{Energy}_i} \tag{3.14}$$

- Energy_i = energy consumed in the i^{th}instance.
- N = total number of samples.

The energy efficiency of the combined three slices has been calculated to measure the total consumption of energy.

A reward function will be evaluated based on these calculations of all the metrics, then these are computed and compared with the previous machine learning techniques that have been used earlier in intelligent resource management in 5G systems.

3.8 Learning and Optimization Loop

This system keeps improving through the continuous learning process. First, the system learns from past experience by storing it in a replay buffer. This makes the system learn from past decisions and improve upon those decisions. Later on, the training is carried out on mini-batches, which helps in learning steadily while minimizing errors. The use of target network helps in maintaining the steady learning process without sudden changes. Federated learning allows different parts of the network to learn independently while sharing just the learned knowledge, instead of all the data.

3.9 Engineering Standards

3GPP TS 38.401 – NG-RAN Architecture

- This document presents the architecture of NG-RAN systems that can be used within 5G networks.
- This document provides the guidelines regarding network slicing, gNB functions, and radio resource management.
- The architecture of the system proposed in this work satisfies the requirements specified in this document through the use of slice-aware resource allocation for eMBB, URLLC, and mMTC.

IEEE P2801 – ML in Wireless Communications

- This standard focuses on the incorporation of machine learning techniques into wireless communication networks.
- The use of AI models such as LSTM, GNNs, and reinforcement learning is supported for

efficient network optimization.

- The suggested model conforms to this standard in that LSTM is used for predicting traffic and Federated DRL for allocating resources.

O-RAN Alliance – RIC (RAN Intelligent Controller) Architecture

- The O-RAN concept focuses on an open approach to the use of radio access networks via the use of AI-based controllers.
- Frameworks have been established in order to maximize the network resources in real time or near-real time manner.
- The suggested solution meets the O-RAN requirements because of its application of AI based decision making (DDQN).

3.10 Multi-disciplinary Aspect

Wireless Communications and 5G:

The project depends on concepts of 5G networks, including network slicing, QoS provisioning, and resource allocation in radios. It is crucial to have a clear idea about traffic types (eMBB, URLLC, and mMTC) and their requirements for designing an effective allocation policy.

Machine Learning and Deep Learning:

In the system case, we use LSTM models to predict traffic behavior in time and GNNs to model network topology. Knowledge about deep learning methods is required for designing these models, especially sequence learning and graph-based learning techniques.

Data Science and Simulation:

The project implies the creation and processing of artificial data sets, analysis of movement of traffic and verification of results on the basis of indicators such as RMSE, MAE, throughput, and latency. The skills that needed includes data cleansing and visualization techniques.

Computer Networks and System Design:

The combination of various models in a single pipeline requires a holistic perspective on networking, resource scheduling, and decision-making. This system is designed to ensure seamless integration between the three processes of prediction, learning, and allocation.

Energy-Efficient and Sustainable Systems:

This system is energy efficient and spectrum-efficient, thereby making it ideal for the sustainability of 5G technologies. The design addresses the current trend in engineering technology of conserving energy without compromising on efficiency.

CHAPTER 4

SOFTWARE DESCRIPTION

4.1 Python

The python is the primary programming language for developing the proposed intelligent radio resource management system for this project. It is widely used for implementing ML techniques due to its simplicity, flexibility, and numerous available libraries. The role of Python in the project involves data preprocessing and feature extraction of the generated 5G traffic dataset. In addition, it allows the development of several ML models, such as LSTM, GNN, and DRL, which are responsible for predicting traffic flow, modeling the network architecture, and making decision-making regarding radio resource allocation. Moreover, Python allows evaluating the performance of the developed system by calculating essential metrics, including Latency, Throughput, Fairness, and Energy Efficiency. Also the plots have been plotted using the libraries for visualization and traffic analysis.

4.1.1 Features of Python

- A user-friendly programming language ideal for rapid prototyping
- Extensive libraries such as NumPy, Pandas, TensorFlow/ PyTorch, and Scikit-learn
- Excellent ML and DL capabilities
- Seamless integration with simulations and data sets
- Efficient data visualization using Matplotlib and Seaborn
- Free and open source, with widespread adoption in the development community

4.2 Google Colab

The Google Colab platform is used for implementing machine learning models and conducting analysis in the scope of this project. It is a cloud-based platform where Python can be utilized. It offers free GPU and TPU access for making the training process faster. In this work, the use of Colab will be involved in cleaning and processing the data, then trained it using LSTM algorithms to predict traffic and performing DRL for resource allocation and optimization. Besides this, the Colab help in experimentation, debugging, and visualization without having to rely on high-end machines. In addition, the collaborative nature of it allows faster changes and modifications of code, which is ideal for the project.

4.2.1 Features of Google Colab

- Cloud-based platform with no need for local installation
- Free access to GPU and TPU for faster model training
- Supports Python and major machine learning libraries
- Easy integration with Google Drive for data storage
- Interactive notebooks for code execution and visualization
- Enables real-time collaboration and sharing

4.3 Ubuntu (VMware Workstation)

This project employs Ubuntu through the use of VMware Workstation to emulate a 5G network and generate fake traffic data. Ubuntu provides a reliable platform that can be used for simulations and setting configurations. Ubuntu creates simulations of various 5G traffic types, such as enhanced mobile broadband (eMBB), ultra-reliable low-latency communication (URLLC), and massive machine-type communications (mMTC). Ubuntu generates real-world data sets for packet arrival rate, traffic volume, maximum latency, and channel status. The data sets are then imported into Python programming for further analysis and modelling. The use of VMware Workstation enables virtualization, meaning Ubuntu can be installed without interfering with other operating systems on the host computer.

4.3.1 Features of Ubuntu (VMware Workstation)

- Stable and open-source Linux environment for simulation.
- A Linux-based open-source simulation environment.
- Supports networking configurations and dataset creation.
- Works well with Python and machine learning applications.
- Suitable for virtualization with the use of VMware Workstation.
- Allows efficient resource allocation for simulations.
- Popular among researchers and developers.

4.4 Machine Learning Libraries

A variety of machine learning and deep learning libraries are employed to develop predictive models in this study. Libraries including TensorFlow and PyTorch are employed to construct LSTM and DRL models, respectively, while scikit-learn library is employed to preprocess and analyze data. NumPy and Pandas libraries facilitate numerical computation and data management, thereby making it easier to manipulate large sets of data.

4.4.1 Features of ML Libraries

- TensorFlow and PyTorch can be utilized for developing deep learning frameworks like LSTM for predicting traffic flow, GNN for spatial modeling, and DDQN (Double Deep Q-Network) for decision-making through reinforcement learning.

- The frameworks also facilitate federated deep reinforcement learning (FDRL), which allows for distributed training with multiple agents that learn independently but do not exchange their raw data.

- Scikit learn is used in data preprocessing, feature scaling, and accuracy by getting RMSE and MAE values.
- NumPy enables efficient numerical computations and matrix operations required for model training and optimization.
- Pandas is used for data handling, dataset creation, and analysis of 5G traffic features.
- These libraries support efficient processing of large-scale datasets and scalable AI model deployment, along with strong community support and documentation.

4.5 ML Algorithms

Algorithm 1. Slice-Aware Federated DDQN Resource Allocation Framework

Initialize:

1: Initialize LSTM traffic prediction model parameters θ_{LSTM}
2: Initialize GNN topology model parameters θ_{GNN}
3: Initialize main DDQN parameters ω^m
4: Initialize target DDQN parameters ω^r
5: Initialize global federated model parameters ω^g
6: Initialize replay buffer D, learning rate α, discount factor λ
7: Initialize maximum training episodes E_{max}, time steps T_{max}, and aggregation rounds K
8: Predict slice traffic demand using LSTM

$$\hat{\lambda}_s(t+1) = f_{LSTM}(H_t)$$

9: Update traffic load feature in state vector
10: Apply GNN to capture spatial interference

$$X' = \sigma(\hat{A}XW)$$

11: Update coordinated multi-cell network features
12: **for** time step $t = 1{:}T_{max}$ **do**
13: **for** each agent $i = 1{:}N$ **do**
14: Observe state s_t^i
15: Select action $a_t^i = \{\phi_{i,k}, P_i\}$ using ε-greedy policy
16: Execute slice-aware scheduling
• URLLC mini-slot priority
• eMBB adaptive allocation
17: **end for**
18: Environment returns reward
19: Observe next state s_{t+1}
20: Store experience tuple (s_t, a_t, r_t, s_{t+1}) into replay buffer D

Algorithm 2. DDQN Training Step

1: Sample mini-batch from replay buffer
2: Compute target value using DDQN:

$$y = r + \lambda Q(s', argmax_{a'} Q(s', a'; \omega^m); \omega^r)$$

3: Update main network weights:

$$\omega^m = \omega^m + \alpha(y - Q(s, a; \omega^m))\nabla Q$$

4: Minimize loss:

$$L(\omega^m) = E[(y - Q(s, a; \omega^m))^2]$$

5: Soft update target network:

$$\omega^r = \tau\omega^m + (1-\tau)\omega^r$$

6: **end for** (time step loop)
7: **end for** (episode loop)
8: **for** round k = 1 to K **do**
9: Each agent uploads local model ω_i
10: Aggregation server computes global model:

$$\omega^g = \frac{\sum_{i=1}^{N} \eta_i\, \omega_i}{\sum_{i=1}^{N} \eta_i}$$

11: Broadcast updated global model to all agents
12: **end for** (federated round loop)

Algorithm 1 describes the overall slice-aware federated DDQN resource allocation process. The model parameters for LSTM, GNN, and DDQN are first initialized. At each time step, the LSTM predicts future traffic demand, while the GNN captures spatial interference and network structure to enhance the system state. Based on this state, each agent selects channel and power allocation actions using an ε-greedy policy. URLLC traffic is given priority, and the left out resources are allocated to eMBB. The system then receives a reward and stores the experience in a replay buffer for training.

Algorithm 2 explains the learning and federated aggregation process. A mini-batch of experiences is sampled from the replay buffer to calculate the DDQN target value and lessen overestimation. To calculate the DDQN target value along with reduce overestimation, a mini-batch of experiences is taken from the replay buffer. While the target network is updated gradually for stability, the main network is updated via temporal-difference loss. Each agent transmits its model parameters to a central serv er following local training, when weighted averaging is used to aggregate them. All agents are then given access to the updated global model, allowing for cooperative and private learning throughout the network.

CHAPTER 5

SIMULATION RESULTS AND INFERENCES

5.1 Simulation Parameters

The proposed **model** was constructed with Python through simulation in a 5G environment. A synthetic dataset representing three major 5G traffic slices - eMBB, URLLC and mMTC, with their Qos metrics were generated and then evaluated. The dataset contained, among other things, the main parameters of latency, throughput, Signal to Interference Ratio (SINR), packet length and queue length. The performance of the LSTM, GNN and Federated Deep Q-Network model as an integrated part was assessed through various simulated scenarios and under dynamic conditions. The results achieved through simulation were evaluated through standard performance metrics to determine whether or not the proposed approach is effective.

The parameters for each model's configuration during training on the dataset are summarized in Table 5.1. The LSTM (Long Short-Term Memory) model is structured with 128 units and a sequence length of 20 in order to better capture temporal patterns of network traffic.

The (GNN) Graph Neural Network is made up of two layers: Layer 1 (32) → Layer 2 (16). These 2 layers will help identify spatial relationships and provide interference characteristics of traffic on the network.

The DDQN (Double Dueling Q-learning) method applies an Epsilon Greedy method with decay for exploring and exploiting during training. The learning rate of 0.001 and discount factor of 0.9 will provide convergence that is both stable and efficient. Each of these models will be implemented in Python using the appropriate libraries, which will provide performance plots and other evaluation metrics. This unique combination of model configurations, implementation and training will allow for accurate learning and optimal performance of the overall system.

Table 5.1: Model simulation parameters

Parameter	Value	Parameter	Value
LSTM Input Features	4	**Learning Rate (DDQN)**	0.001
Sequence Length	20	**Discount Factor (γ)**	0.9
LSTM Units	128	**Exploration Strategy**	ε-greedy
GNN Layers	2 (32 → 16)	**Initial epsilon(ε)**	0.45
Graph Type	k-NN (k = 5)	**Minimum Epsilon (ε)**	0.10
Policy Network (DDQN)	64 → 32 → 3	**Epsilon (ε) Decay**	0.995
Action Space	3	**Training Epochs**	100
State Input	GNN + LSTM	**Batch Size**	16
RMSE	11.21	**MAE**	7.67

5.2 Traffic analysis of the 5G Dataset

The validated dataset values of 5G traffic types are presented in Table 2 below.

- URLLC has the lowest latency (~2.99 ms) and moderate throughput (~85 Mbps); therefore, it is suitable for real-time applications.

- eMBB has the highest throughput (~150 Mbps); however its latency is also relatively high (~35 ms) resulting in eMBB being ideally suited for high-bandwidth services, such as video streaming

.

- mMTC has the lowest throughput (~11 Mbps) among 5G traffic types and the highest latency (~54 ms) thereby making mMTC adequate for IoT based applications that do not have strict timing requirements.

The realistic 5G behavior shown by the dataset demonstrates that each slice has different characteristics for QoS and validates that the model was developed based on realistic and useful data.

Table 5.2: Validated dataset with all 3 slices

Traffic Type	Latency(ms)	Throughput (mbps)	SINR (dB)
URLLC	2.994360	85.459700	25.003739
eMBB	35.817932	150.653918	19.994957
mMTC	54.538996	11.972889	9.965159

The features were mentioned in Chapter 3 and from this we could also represent the data through the form of boxplot where it accurately follows the QoS differences and realistic 5G traffic trend as:

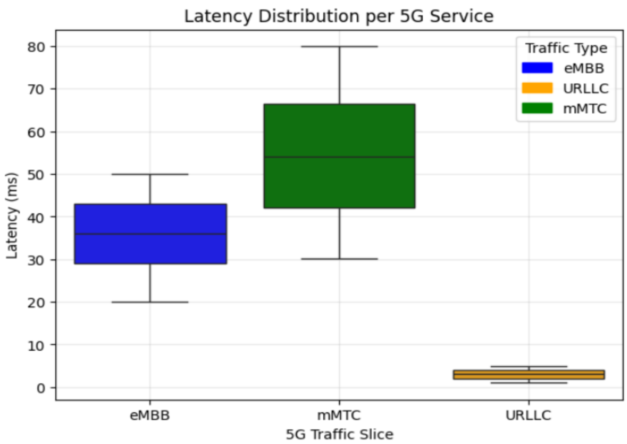

Figure 5.1: Latency distribution per slice

The latency distribution plot shows how the delay varies by 5G traffic slices. From this, it is clear that the lowest latency with the least amount of variation is produced by ultra-reliable low latency communication; these are critical for real-time, delayed-sensitive applications. Enhanced mobile broadband (eMBB) will produce moderate levels of latency (speed vs. data), while massive machine-type communications (mMTC) have the highest latencies but this is expected since the applications are non-time critical (IoT-based communication).

The plot shows the service was successfully able to provide QoS differentiation across all traffic classes.

5.3 Dataset Validation using ML Models

Table 3 illustrates how LSTM and Random Forest models performed in generating and public datasets utilising RMSE and MAE as evaluation metrics. The LSTM (Long Short-Term Memory) model is a deep learning method for making time-series forecasts, which can learn sequential data like network traffic and other types of temporal data. Random Forest, however, uses many decision trees to create machine-learning methods (either regression or classification), but it does not capture temporality as well as an LSTM would.

To evaluate model performance, two error metrics are used. RMSE computes the square root of the average squared difference between predicted and actual values, thus giving higher importance to larger errors:

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2} \qquad (5.1)$$

Table 5.3: Comparison between public and generated dataset

Model	**Dataset**	**RMSE**	**MAE**
LSTM	Generated	12.69	10.84
	Public	32.97	27.78
Random Forest	Generated	11.02	7.53
	Public	22.29	19.16

MAE measures the average absolute difference between predicted and actual values which in turn gives a measure of prediction accuracy:

$$MAE = \frac{1}{N}\sum_{i=1}^{N} \mid y_i - \hat{y}_i \mid \qquad (5.2)$$

The lower error rates of the generated dataset (LSTM RMSE ≈ 12.69; MAE ≈ 10.84) compared to the public dataset indicate that the public dataset is a much more complex and noisier dataset than the generated dataset by means of being more stable and having a more organized structure, allowing the models to learn better.

Therefore, this affirms the accuracy of the generated data in terms of traffic prediction and the efficiency in allocating resources for the new traffic forecast model.

5.4 Traffic Data Analysis

5.4.1 Traffic Burstiness Plot

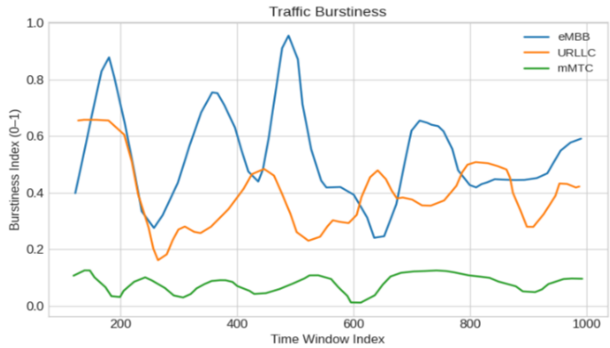

Figure 5.2: 5G traffic burstiness of the 3 slices

Through the variability of traffic patterns over time for various 5G slices, the burstiness plot in figure 5.2 provides visibility into the extent of these variations. While eMBB shows high burstiness due to being a large-scale application that utilizes a lot of bandwidth, URLLC has a moderate amount of variability and will have consistent traffic patterns. Conversely, mMTC has low levels of variability and is generally stable and predictable in the utilization of resources by connected devices. The burstiness plots confirm that the traffic behavior diversity is captured well by the model, allowing for intelligence and adaptive allocation during periods of demand

5.4.2. Data Rate Plot

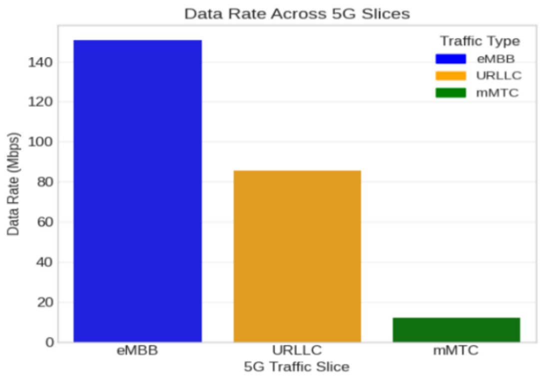

Figure 5.3: Data rate across 5G slices

The Data Rates Graph in Figure 5.3 presents how data transfers vary based on the distribution of 5G Traffic in different slices. Data rates for eMBB slices are the highest; eMBB slices were designed to provide high data transfer rates specifically for high data volume applications such as video streaming and multimedia services. The URLLC slice data rate provides moderate-rate access while maintaining reliability and low latency for real-time communication applications. The mMTC slice will have the lowest-rate access as it is only designed for low-power IoT devices that need very little data transmitted. The Data Rates Graph shows that the differences between data rates across slices correspond to the design goals for each slice type, further confirming that resource allocation based on a slice awareness strategy works well.

5.4.3. Packet Arrival Rate Analysis

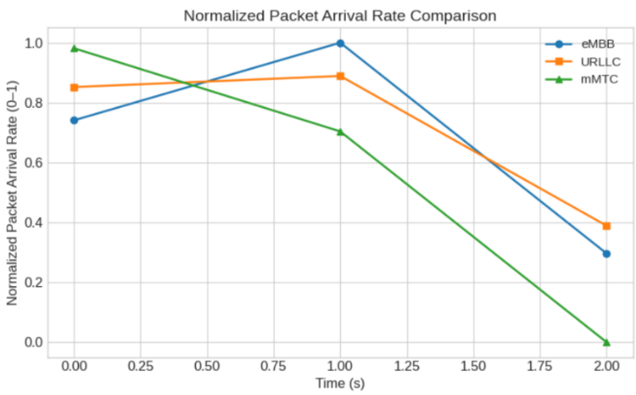

Figure 5.4: Packet arrival rate across 5G slices

Figure 5.4 presents the plot of packet arrival rates over time for each slice in 5G (i.e., mobile network) technology. The following observations can be made based on these trends: the dynamic nature of traffic within the eMBB slice is evident through the initial periods of increasing volume spikes, followed by rapid reductions. However, traffic associated with maintains a relatively stable level of packet volume throughout this same period, while mMTC gradually and consistently reduces through the time period, which corresponds to its typical low volume and inconsistent behavior in IoT devices. Ultimately, this data illustrates the differing nature of the various 5G network slices and emphasizes the necessity for adaptive resource allocation among these networks.

5.5 GNN Embeddings PCA scatter plot

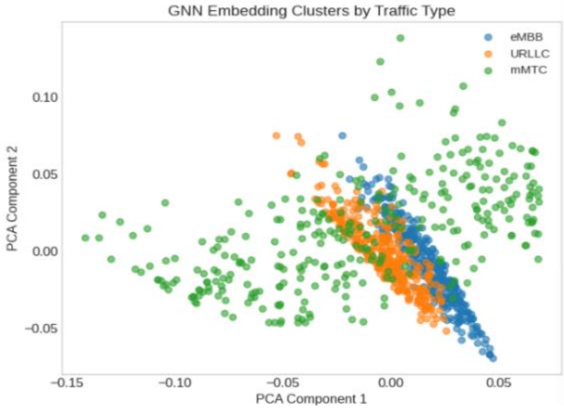

Figure 5.5: GNN scatter plot of the 3 slice

A scatter plot (refer to Figure 5.5) created using PCA displays the different traffic type-generated GNN embeddings on the x-axis and y-axis as their reduced feature components. It is evident from the scatter plot that the eMBB, URLLC, and mMTC slices are clustered separately, showing different traffic characteristics for each type of slice. Results show that eMBB and URLLC points are relatively close together because of their similarly moderate-to-high traffic, while mMTC points are more dispersed throughout with a varying amount of irregular and diverse traffic pattern behaviour. The clustering of all three types of traffic supports the efficiency of the model at distinguishing between differing 5G traffic patterns.

5.6 LSTM Throughput Prediction plot

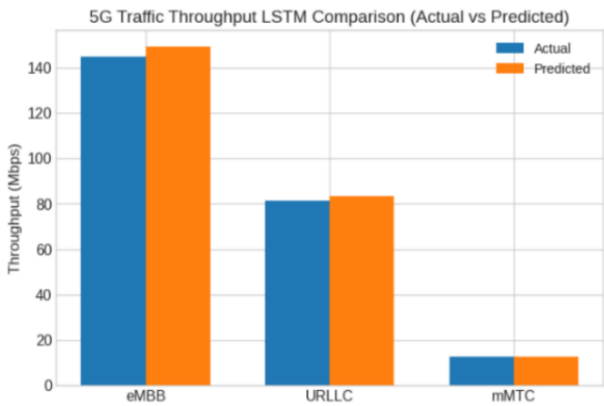

Figure 5.6: LSTM prediction of 5G slices throughput

As shown in Figure 12, the predicted LSTM throughput values are very close to the corresponding actual throughput values for each of the 5G traffic types. The predicted eMBB throughput value (~150 Mbps) is nearly equal to the actual eMBB throughput value (~145 Mbps), so the model has been able to learn patterns with respect to the high bandwidth components of traffic. The predicted ULRC throughput value (~84 Mbps) also closely resembles the actual ULRC throughput value (~82 Mbps) for the given data. The model has also been accurate in the case of the predicted mMTC throughput value (~13 Mbps) for which the actual mMTC throughput value (~12 Mbps) is very close to the predicted mMTC throughput value. Overall, the small difference among these actual and predicted throughput values indicates that the LSTM model has the ability to learn temporal patterns while producing reliable predictions of throughput for 5G traffic.

5.7 Model performance and comparison table

The performance metrics in this comparison include throughput, latency, packet loss, spectral efficiency, and energy efficiency. These are described in detail in chapter 3. These performance metrics are critical in assessing the effectiveness of the proposed system by quantifying data transmission efficiency, delay performance, reliability, bandwidth utilization and energy consumption. Collectively, they provide a complete picture of the performance of the model when operating under various network conditions. Table 4 presents the comparison of the framework incorporating LSTM-based prediction with FDRL and DDQN algorithms to basic DRL and dueling DQN approaches.

Table 5.4: Performance comparison table

Parameters	**Basic DRL**	**Dueling DQN**	**Proposed (LSTM+FDRL+DDQN)**
Avg Throughput (mbps)	50.18	60.22	**83.94**
Avg latency (ms)	60.22	41.5	**30.84**
Packet Loss	51%	36%	**19%**
Spectral Efficiency (bps/Hz)	1.07	1.37	**1.78**
Energy Efficiency (Btu/h per Watt)	0.85	1.1	**1.41**

The average throughput of the proposed model, which is comprised of three 5G slices, is 83.84 Mbps and demonstrates outstanding performance when compared to Basic DRL and Dueling DQN. The average latency (30.84 ms) and packet loss (19%) show that the proposed model provides superior QoS support for the combined slices. In terms of both latency and packet loss, Basic DRL has the worst case; therefore, it is not effective when scheduling resources. The proposed model also has the highest spectral (1.78 bps/Hz) and energy (1.41 Btu/h/Watt) efficiencies, demonstrating good utilization of bandwidth and distribution of resources across all three 5G slices.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This simulation results gives an intelligent radio resource management framework for slicing in next-generation 5G networks. The framework supports 3 traffic types, namely; eMBB, URLLC, and mMTC. The proposed solution generates a synthetic dataset with higher accuracy than existing public datasets compared to published analysis. This enabled the use of synthetic datasets in creating effective models for training and validating traffic management network. The proposed framework combines an LSTM traffic prediction algorithm and FDRL using a DDQN agent to dynamically allocate and distribute radio resources. After analysis of results from simulations based on FDRL radio resource management framework demonstrated improved performance compared to traditional machine learning (ML) approaches with approximately 83.94 Mbps throughput, approximately 30.84ms latency, and approximately 19% fewer packets dropped than basic ML models. Furthermore, the use of this framework also enhanced spectral (1.78 bps/Hz) and energy efficiency (1.41) thus providing optimal use of the radio resources available within the 5G network. The proposed intelligent radio resource management framework provides a scalable, efficient, and intelligent solution for future 5G networks and beyond.

6.2 Future Work

The Further Development of 5G and Beyond, The Proposed Framework Shall Be Implemented Through:

- The Extension of The Model to Real-Time Deployments (5G/6G) And Actual Live Traffic data.
- The Incorporation of More Advanced Artificial Intelligence Models Such as Transformer- Based Architectures to Provide Better Predictions of Conditions.
- Enhancing Federated Learning Algorithms Using Secure Aggregation and Privacy Protection Methods.

- Optimizing Systems for Edge Computing and IoT Applications.
- Incorporating Mobility-Aware Resource Allocation for Dynamically Moving Users.
- Applications of the framework will be developed for emerging applications such as Smart Cities, Autonomous Vehicles, And Healthcare Systems in the current coming future 5G world.

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9% detected as AI

AI detection includes the possibility of false positives. Although some text in this submission is likely AI generated, scores below the 20% threshold are not surfaced because they have a higher likelihood of false positives.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.